

- 2 Golden orb weaver spiders, *Nephila pilipes*, are found in East Asia, South-east Asia and Australia. They are active in the day and at night. Golden orb weaver spiders build webs in trees and shrubs to catch different species of insects for food.

All female golden orb weaver spiders are black with yellow spots on their legs and body. Biologists think that this adaptation evolved due to natural selection. Spiders with yellow spots may be able to attract more insects to their webs.

Fig. 2.1 shows a female golden orb weaver spider sitting on a web.



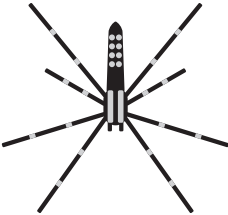
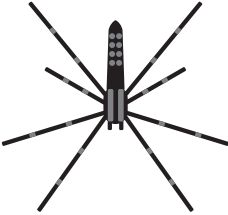
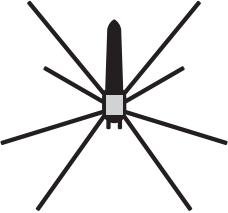
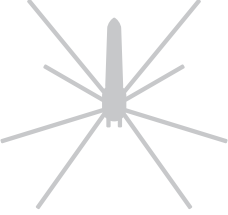
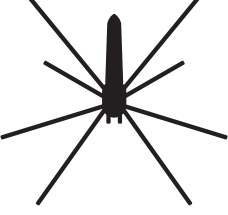
Fig. 2.1

Some biologists visited a forest in East Asia in July 2008. The biologists found five webs that were approximately the same size. Each web belonged to a female golden orb weaver spider.

The biologists decided to investigate how the colour and pattern of spots on the spiders affect the number of insects attracted per hour to each web (insect attraction rate).

The biologists made two-dimensional (2D) models of spiders, as shown in Table 2.1.

Table 2.1

model	description	diagram
A	- black body with yellow spots - black legs with yellow spots - same colour and pattern of spots as golden orb weaver spiders	
B	- black body with blue spots - black legs with blue spots - same pattern of spots as golden orb weaver spiders	
C	- black body with large yellow spot - black legs - total area of yellow colour is the same as model A	
D	- yellow body - yellow legs	
E	- black body - black legs	



At the first web, the biologists:

- removed the living golden orb weaver spider from the web
- placed model **A** in the centre of the web
- placed a video camera 1 m from the web
- filmed the web and model for 6 hours
- recorded an insect attraction event whenever an insect flew towards the model, touched the model, or touched the web
- calculated the insect attraction rate of model **A**.

The procedure was repeated by placing models **B**, **C**, **D** and **E** on the four other webs.

The whole investigation was repeated 25 times, and a mean insect attraction rate for each model was calculated.

(a) (i) Identify the **dependent** variable in this investigation.

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(ii) The pattern of spots on models **A** and **B** was the same as the pattern of spots on female golden orb weaver spiders.

Identify **two other** variables that the biologists should standardise when making the models for this investigation.

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(iii) Suggest a suitable control for this investigation.

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(b) Fig. 2.2 shows the mean insect attraction rates for the five models. The error bars show $\pm$ one standard error (SE).

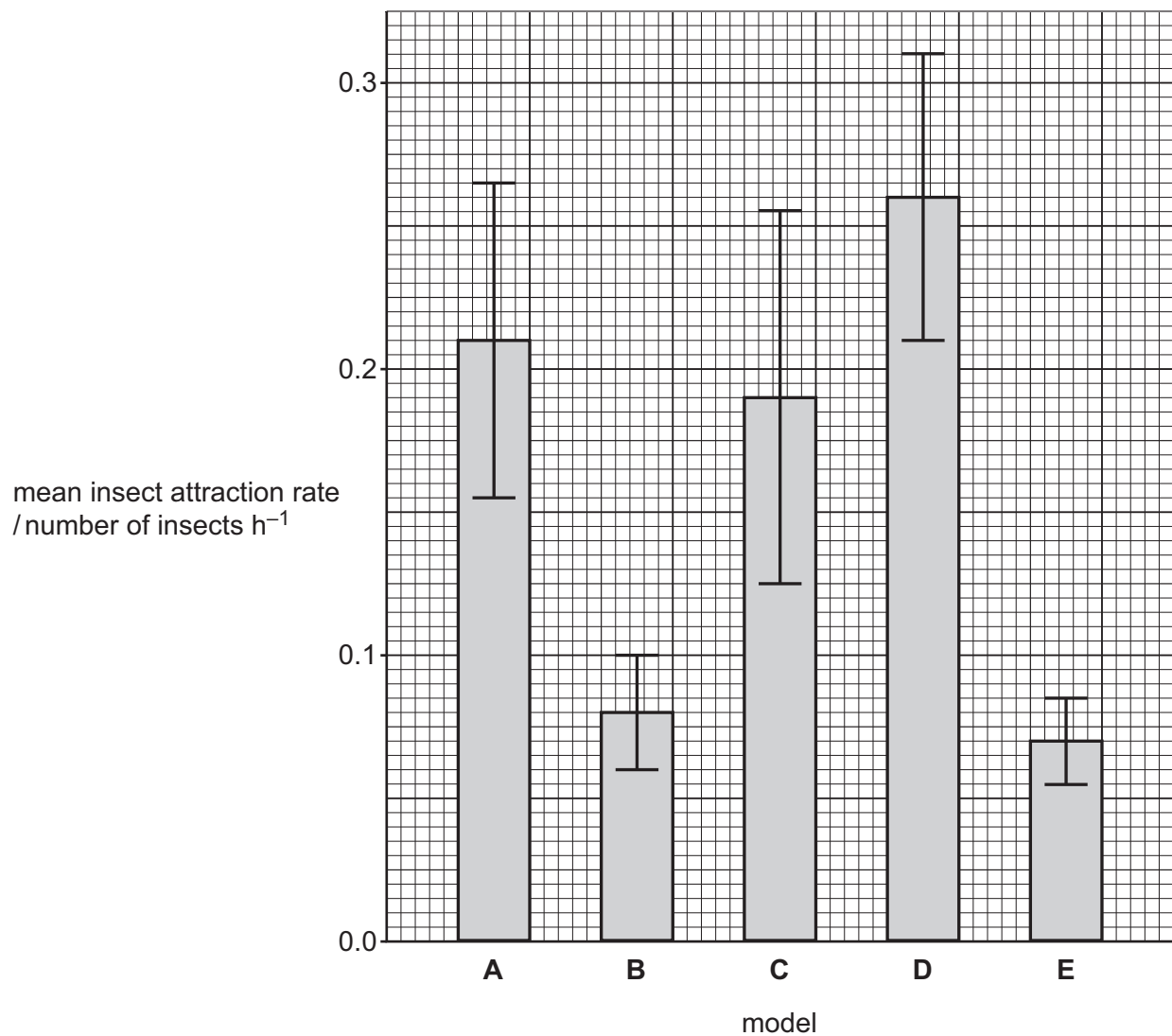


Fig. 2.2

- (i) Use the information shown in Table 2.1 and Fig. 2.2 to discuss the conclusions that can be made about the effect of colour **and** pattern of spots on the models on insect attraction rates.

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(ii) The biologists decided to analyse the data in Fig. 2.2 using t -tests.

State a null hypothesis the biologists could make before carrying out the t -tests.

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(iii) State **two** reasons why the t -test is suitable for analysing the data shown in Fig. 2.2.

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(c) A student thought that the investigation did not provide enough information about how the colour and pattern of spots on female golden orb weaver spiders help to attract insects to their webs.

Suggest how this investigation could be improved to increase confidence in the results.

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[Total: 13]